SON 2337: National Framework for Environmental Pool Management: A Standardized Approach to Prioritizing Ecological Operations at USACE Reservoirs

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Problem:

USACE manages approximately 700 dams where environmental pool management could deliver ecological benefits through minor operational adjustments, but the agency lacks a standardized methodology for identifying which reservoirs are best suited. Implementation has been opportunistic rather than systematic. A Sustainable Rivers Program pilot study successfully demonstrated a screening methodology at five Kansas City District reservoirs, but the approach needs to be generalized and scaled for national application.

Envisioned Product:

An EPM Decision Support Toolkit provides a scalable solution for reservoir management, featuring an extended and independently validated regional case study, a National Prioritization Tool for ranking reservoirs with nationally available data, a Standardized Methodology Report that documents scoring criteria for various environmental factors, and Implementation Guidance complete with templates for deviation requests and monitoring plans.

Value Added:

This project transitions environmental pool management from ad hoc, project-specific efforts to a standardized, data-driven practice applicable across the full Civil Works portfolio. By providing a systematic framework for identifying where minor operational adjustments yield the greatest ecological benefit, the agency can direct limited study resources efficiently and generate the technical justification Districts need to initiate Water Control Manual updates, reducing the time and cost of the study process. This enables operating at both the macro level, to prioritize reservoirs and resources, and at the micro level, to enhance site-specific environmental performance, protection, and operational efficiency.

Regional/National Significance:

The framework applies to all USACE Districts and Divisions managing reservoir systems. The prioritization tool uses only nationally available datasets (Sentinel-2, Landsat, Global Surface Water Dataset, National Land Cover Database, USACE Access to Water), ensuring immediate applicability without District-specific data collection. The products provide the scientific basis for Planning studies and Water Control Manual updates under ER 1110-2-240, support proactive compliance with the Endangered Species Act and Clean Water Act, and advance the USACE Environmental Operating Principles. EPM enables ecological gains from existing infrastructure with no construction costs.